

flamefast

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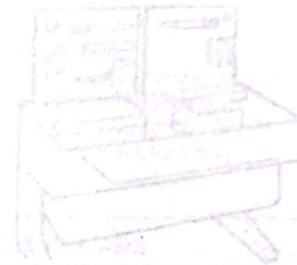
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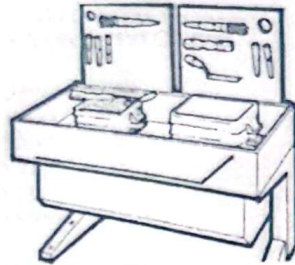


TOOLS · MATERIALS &
EQUIPMENT USED FOR

MOULDING BENCH

Has a metal lined working top surface upon which moulding flasks are placed for ramming and working. The bench top also has a recessed area to store moulding sand.

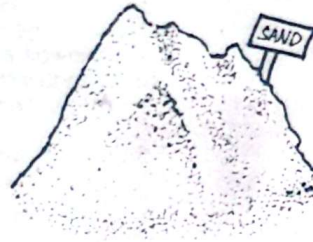
1



MOULDING SAND

Is used to form the moulding cavity and must be capable of withstanding the heat of the molten metal. The finest sand, called facing sand, is used next to the pattern to form a good surface finish to the mould cavity. Green sand is used to fill the main volume of the mould.

2

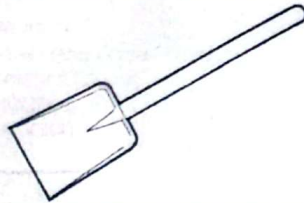


MOULDERS SAND SHOVEL

Materials - Lightweight steel,
Square dished blade
with wooden handle.

Use for scooping, sluicing and tempering the sand and for filling the flask with moulding sand.

3



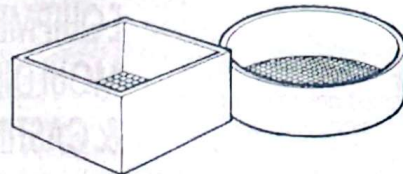
FOUNDRY RIDDLE

Materials - Wood frame with
wire mesh.

Use A - FINE RIDDLE has $\frac{1}{16}$ in square holes in the mesh to produce the fine facing sand for use next to the pattern.

Use B - COURSE RIDDLE has $\frac{1}{4}$ in - $\frac{3}{8}$ in square mesh and is used to aerate the sand to give permeability and also clean any contamination from the sand.

4



BOTTOM BOARD

Material - $\frac{3}{4}$ - 1in thick plywood.
USE - The Bottom Board is placed on the Moulding Bench and supports the Moulding Flask.

5

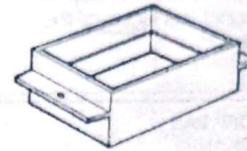
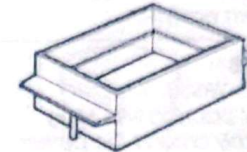


MOULDING FLASKS

Materials - Wood, sheet metal,
Aluminium, Grey Iron.
Consists of 2 sections.
The bottom half is called the 'Drag' and the top half is called the 'Cope'.

USE - These are the containers which hold the sand when making the mould.

6

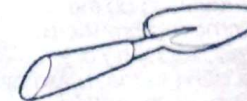


PATTERNS

Materials - Wood, Metal or
Plaster of Paris if they are to be permanent, or can be made of Wax, Plastic, Resins or Polystyrene, where the pattern is destroyed by the hot cast metal.

USE - To provide the shape of the mould cavity in the sand. Patterns should be oversized to allow for shrinkage and machining. They also can be slightly tapered for ease of removal from the moulding sand.

7

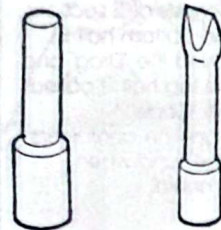


PARTING DUST

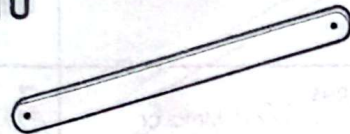
Materials - Fine dry sand, brick dust or fine charcoal.
 USE - Parting Dust is placed in a coarse weave bag and is used on the surface of the moulding sand so that the cope and drag can be separated easily, and so that the pattern can be withdrawn easily.

8**RAMMERS**

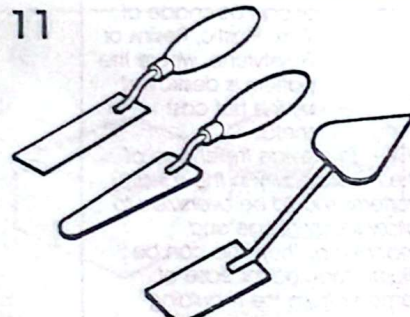
Material - Wood.
 USE - For packing Moulding Sand firmly around the pattern and Sprue Pins. The wedge shape type is used in corners and around the base of the Sprue Pins.

9**STRIKE OFF BAR**

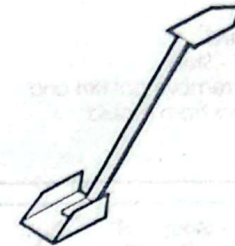
Material - Steel.
 USE - To scrape off the surplus sand and to obtain a smooth level surface on the bottom of the Drag and on the top of the Cope.

10**MOULDING TROWEL**

Materials - Steel blade with wooden handles.
 USE - For touching up the mould, removing any sharp corners and repairing any damage done to the mould by the removal of the pattern.

11**GATECUTTER AND SQUARE**

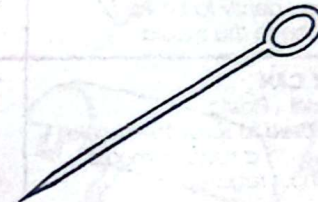
Material - Steel.
 USE - A GATECUTTER is used to cut the channels, called gates, which are run from the mould cavity to the Sprue.
 B - SPOON for cutting the pouring basin.

12**SPRUE PINS**

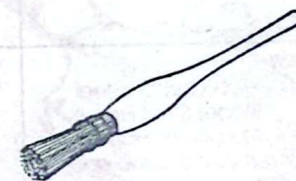
Material - Wood
 USE - The pins are tapered and are used to form a cavity or Sprue in the cope through which the metal will flow to the mould. They can also be used for forming a riser at the opposite end of the mould.

13**VENT ROD**

Material - Steel Piano Wire.
 USE - The Vent Rod is pushed through the Cope or Drag to within 1/4 - 3/8 in of the pattern leaving a number of small passages through which unwanted gasses can be released from the mould.

14**WET BRUSH**

Materials - Wood or Plastic Handle, soft Nylon or Animal Hair Bristles.
 USE - Damping the moulding sand around the edge of the pattern, this helps to prevent the moulding sand crumbling at the edge when the pattern is removed.

15

DRAW PINS

Material - Steel

USE - To remove pattern and Sprue Pins from mould.

16

**RAPPER**

Material - Steel.

USE - When Draw Pins have been screwed into the pattern or Sprue Pin, they are gently tapped in several directions with the rapper to free the Sprue Pin or pattern before lifting out of the moulding sand.

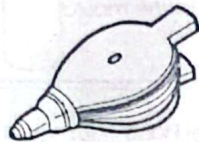
17

**BELLOWS**

Materials - Leather and Wood.

USE - To remove loose sand from the mould cavity and the sprue and riser holes. They must be used gently to prevent damage to the mould.

18

**SPRAY CAN**

Material - Plastic.

USE - Used to spray the mould cavity with a suitable mould coating if required.

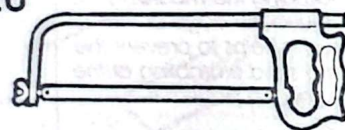
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**HACKSAW**

Materials - High Carbon Steel Blade, Steel Frame, Aluminium Handle.

USE - For removing sprue and riser from the rough casting.

20



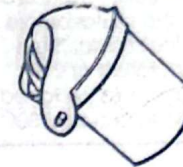
flamefast SAFETYWEAR

The use of Safety apparel in a school workshop cannot be over emphasised, particularly in the Casting or Metal Melting area where students must also be protected from heat, burns and splashed molten metal.

EYE GLASSES AND FACE SHIELD

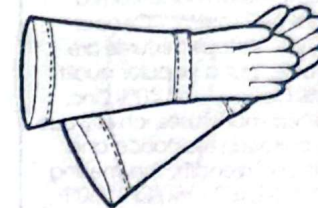
Eye Glasses should be worn at all times, but where a student is involved in charging, skimming, fluxing and pouring metal, he or she, must wear a face shield.

21

**HEAT RESISTANT GLOVES**

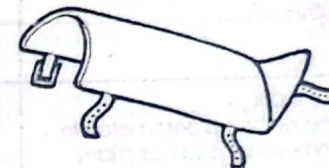
Should extend beyond the elbow and be worn to give protection when charging or drawing the crucible.

22

**KNEE LEGGINGS**

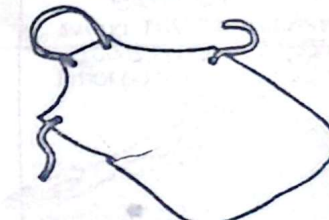
Must be heat resistant and should protect from the knee down and cover the upper part of the shoe.

23

**BIB TYPE APRONS**

Should be heat resistant and must be worn by students handling molten metal to give protection against metal splashes and heat.

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ALUMINIUM

A relatively soft metal but when alloyed with other metals it can be given great strength and still remain very light in weight. The melting point of aluminium and aluminium alloys is fairly low - around 1220°F. But it is necessary to superheat the metal before pouring into the mould, therefore the temperature of the molten metal should be raised in the crucible to around 1300°F.

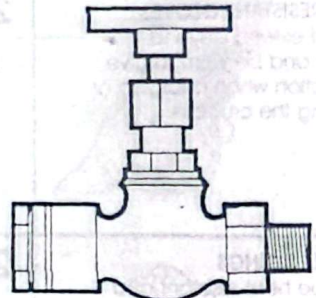
25



BRASS

A yellow metal made from a mixture of 2 metals - Copper and Zinc. Various mixtures are available, but a popular quality is 60% Copper and 40% Zinc. Brass has many uses, chiefly due to its corrosion resistance and moderate strength. The melting point of Brass is around 1860°F, but the metal should be heated to around 1920°F. before pouring.

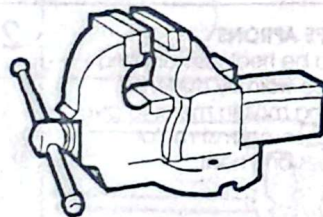
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GREY IRON

Is one of the hardest metals in common use and has many applications in industry. Grey Iron melts at a white heat temperature of 2270°F, but will be poured at around 2600°F, and great care must be taken.

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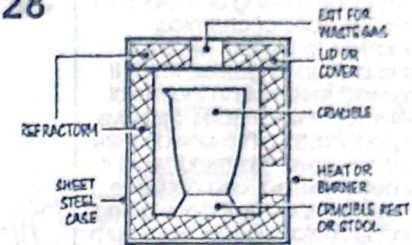


CRUCIBLE FURNACE

Materials - Sheet Steel case with Heat Resistant Refractory lining.

USE - The crucible furnace houses the crucible which contains the metal to be melted to which heat is applied from various fuels, such as Gas, Oil or Solid Fuel.

28



CRUCIBLE

Materials - Clay Graphite or Silicon Carbide.

USE - For holding the metal whilst it is transferred from the crucible furnace to the mould. All crucibles have pouring lip to assist the pouring of the molten metal and they are constructed from materials that withstand the heat of the furnace but rapidly conduct heat from the furnace to the metal.

29



CRUCIBLE REST OR STOOL

Material - Refractory.

USE - To position the crucible at the correct height in the combustion chamber and to prevent the crucible from contacting the cooler furnace floor.

30

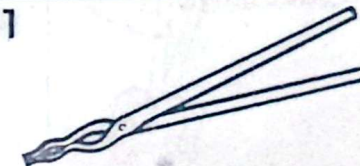


METAL PICK-UP TONGS

Material - Steel.

USE - For handling Hot Metal when charging the furnace or handling rough casting before they have completely cooled.

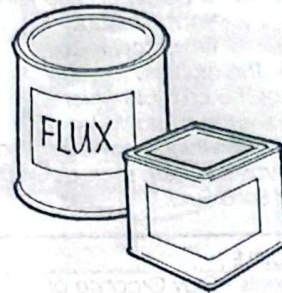
31



CHEMICAL ADDITIVES

During the melting of metal there are two types of chemical additives to prevent oxidation and pick up of gasses, which if allowed to remain in the metal will result in small holes and give a poor casting. Flux powders are used to cover the metal whilst it is being melted, and combine with the impurities to form a slag on the metal surface. Aluminium and Aluminium alloys are extremely prone to dissolving hydrogen and so an additional treatment is required. This is called 'Degassing' and is done just prior to pouring the metal by plunging a tableted chemical below the surface of the metal.

32

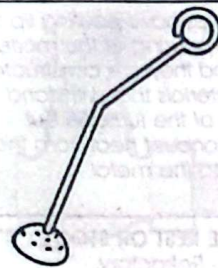


DEGASSING PLUNGER

Material - Steel.

USE - To push the degassing table down into the melt and to hold it there until it has stopped reacting.

33

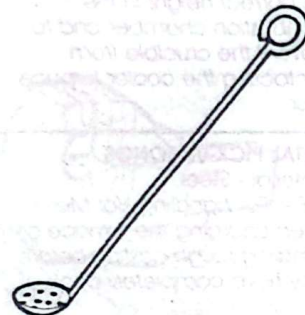


COMBINED DROSS LIFTER AND SKIMMING LADLE

Material - Steel.

USE - For the removal of the slag (impurities) on the top of the metal before pouring.

34



COMBINED PORTABLE THERMOCOUPLE AND PYROMETER

Used to measure the temperature of the molten metal to ensure that it is at the correct temperature for pouring. The thermocouple is the sensing section which is immersed in the molten metal. This is connected to the pyrometer which is simply an instrument for indicating temperature.

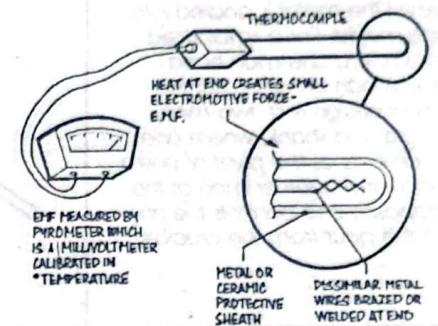
Different wires are used for the thermocouple depending upon the range of temperatures that are required.

Iron-Constant between 600°F. and 1600°F.

Chrome-Alumel between 1200°F. and 2000°F.

Platinum-Rhodium between 2000°F. and 2800°F.

35



600 ————— 1600°F

1200 ————— 2000°F

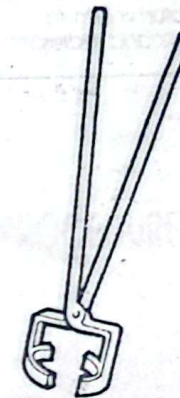
2000 ————— 2800°F

CRUCIBLE LIFTING TONGS

Material - Steel.

USE - For lifting the crucible from the furnace and transferring to the pouring shank. They are designed to grip the crucible securely and must be accurately sized for each size of crucible being used, so that they are closed and grip the hot crucible containing molten metal. They should impart an even pressure and avoid creating stress that might damage or crack the crucible.

36

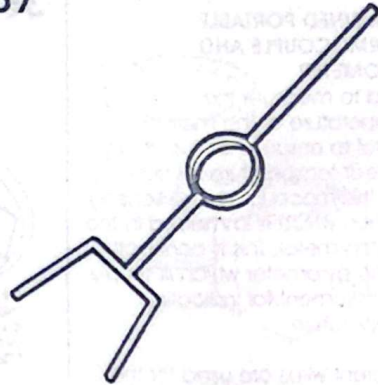


LADLE CARRIER

Material - Steel.

USE - For holding the crucible whilst the metal is poured into the mould. For a small sized crucible a one-man hand shank can be used. But the usual design is a two-man single end shank, where one man acts as the pivot at one end, and another man at the opposite end controls the angle of the pour from the crucible.

37



CERAMIC WASH

Materials - Ceramic Powder.

USE - Mixed with water, the wash is applied or painted on all tools that are likely to come in contact with the molten metal. The wash helps to retard oxidation scaling, increase tool life, prevent iron pick-up contamination and loosen the tendency of aluminium to adhere to tools and ladles.

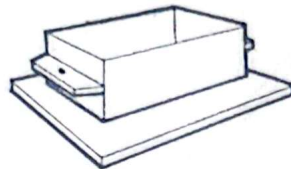
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MOULDING PROCEDURE.

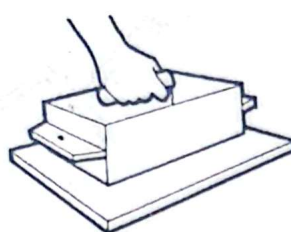
Place the Drag upside down on a Bottom Board. remember, the Drag has to receive the Cope once it has been filled with sand.

1



Place the pattern into the Drag. Its position in the drag depends on the shape of the pattern and where the Sprue Pins will be placed later on. Make sure the Flask is large enough to allow a 2in wall of sand around the pattern.

2



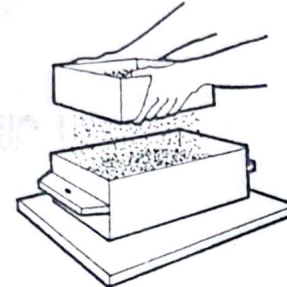
Sprinkle a little Parting Dust on the Bottom Board and the pattern.

3



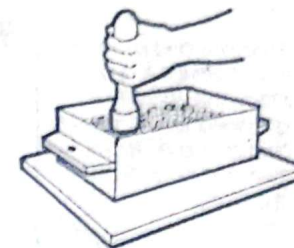
Shake the Facing Sand through a Fine Riddle over the pattern until it is completely covered to a depth of 3/4in.

4



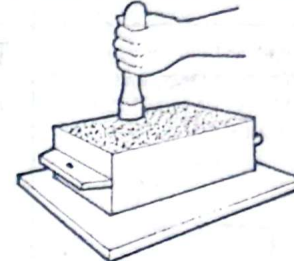
Use a Rammer to compact the Moulding Sand firmly, taking care not to damage the pattern. Remember this sand is very fine and compresses easily. It is recommended to work around the outside of the pattern first so as to prevent it moving. Roughen up the surface to form a key for the next layer of sand.

5



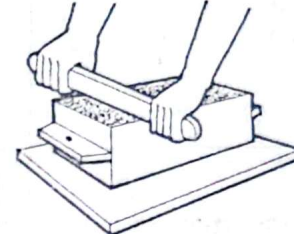
Using a coarse Riddle and Rammer, work in stages until the Flask is full. Rammed sand should be firm but not too hard.

6



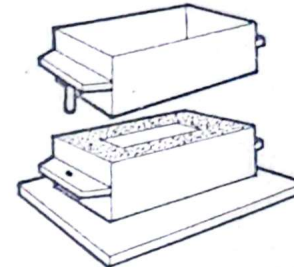
Use a Strike Off Bar to scrape off the surplus sand and leave a flat level surface.

7



Invert the Drag and carefully place it on a clean Bottom Board which has been lightly sprinkled with parting Dust.

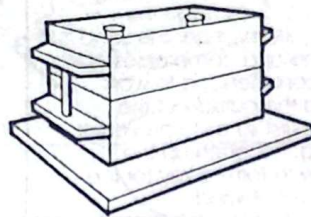
8



Place the Cope in position.

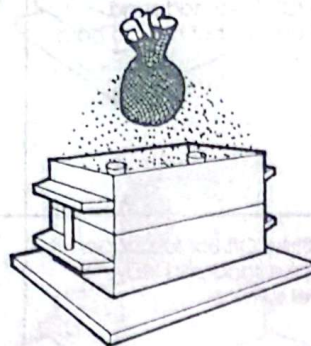
Position Sprue Pins and second part of the pattern if it is a two piece pattern. The Sprue Pins should be placed to suit the pouring arrangements. Avoid placing them too near the edge of the Flask.

9



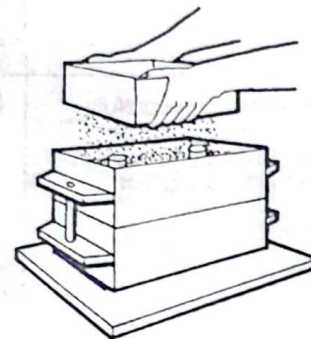
Sprinkle Parting Dust on the jointing face and the now upward facing sand surface. Also lightly dust the pattern.

10



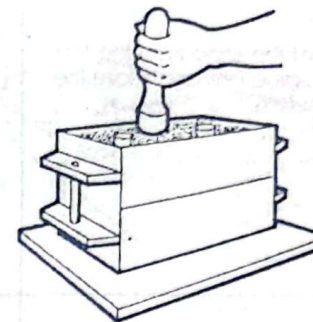
Shake Facing Sand through a fine Riddle until the pattern is covered to a depth of $\frac{3}{4}$ in.

11



Use a Rammer to compact the Moulding Sand firmly. Paying particular attention around the bases of the Sprue Pins.

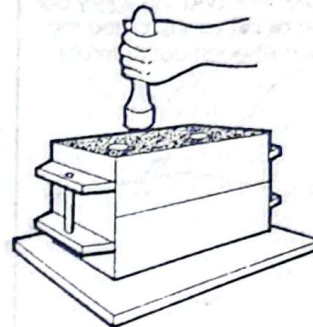
12



Using a coarse Riddle and Rammer work in stages until the Flask is full.

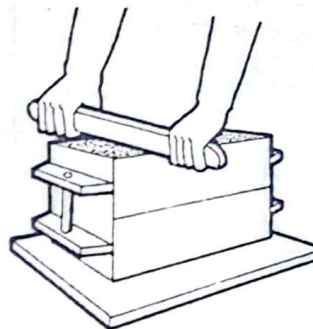
Do Not over Ram.

13



Use a Strike Off Bar to remove the surplus sand from the top of the Cope.

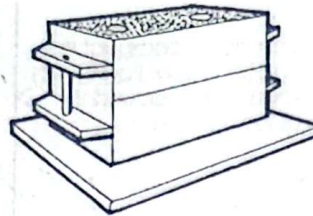
14



Then using the Vent Rod make several holes in the sand to within $\frac{1}{4}$ in of the pattern to allow unwanted gasses to escape.

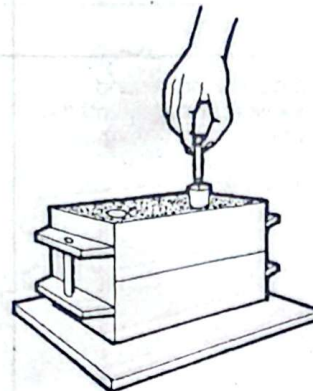
Remove the sand from the top of the Sprue Pins and make the surface firm.

15



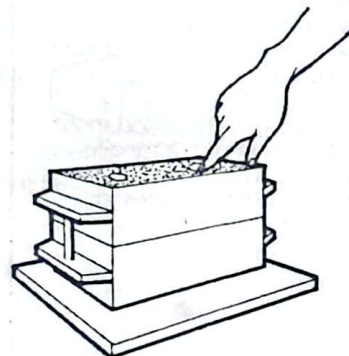
Insert the Draw Pins and Rapping Screws into the tops of the Sprue Pins and rap lightly but firmly in all directions to free the Pins then lift each out carefully.

16



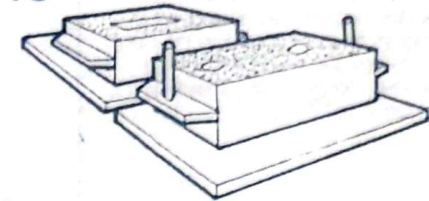
Using the finger, round off the corners carefully. Making the sand quite firm. This is to prevent sand being washed down the hole when the metal is poured.

17



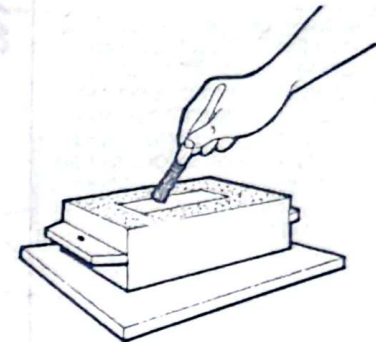
Carefully lift the Cope from the Drag, and place it upside down on a clean Bottom Board which has been sprinkled with Parting Dust.

18



Use a Wet Brush to dampen the sand around the edge of the pattern in the Drag.

19



Insert Draw Pins into the pattern and rap carefully in all directions to free the pattern. Lift the pattern out very carefully from the Drag taking care not to damage the mould.

20

